Pharmacovigilance Follow-Up Solution

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Problem Statement 2

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PROBLEM STATEMENT

Pharmacovigilance is essential for ensuring patient safety by monitoring and reporting the adverse events (AE's) associated with consumed medicines of mentioned dosage. Regulatory guidelines require pharmaceutical companies to collect complete and accurate follow up information when initial AE reports are incomplete and compile them.

In practice, most AE reports lack critical details such as severity, timing, outcome or dosage. Follow-up data collection is often unsuccessful due to complex and time taking process for filling up forms, repeated manual follow-ups, language barriers, limited availability of healthcare professionals and growing concerns around privacy and trust issues.

Existing pharmacovigilance workflows rely heavily on manual processes and static reporting formats, leading to low response rates, delayed case completion, inconsistent data quality and reduced ability to detect emerging safety signals, especially in global and multi country settings where each country has different standards of reporting.

There is a need for a practical, scalable and compliant solution that simplifies follow-up data collection, minimizes respondent burden, improves data completeness along with support of early safety signal detection while maintaining regulatory and privacy requirements.

PROPOSED SOLUTION

In this context, we would like to introduce our proposed AI-assisted pharmacovigilance follow-up solution that aims to enhance both the quality and timeliness of adverse event data gathering while ensuring that it is delivered within the compliance framework.

The proposed solution relies on a consensual and conversational process for taking intakes (for example, using a WhatsApp chatbot) to facilitate the submission of adverse events by patients and medical professionals. This solution enables users to communicate in the desired language. In other words, it provides multilingual interaction.

Upon receipt of the initial report, the use of context-aware AI agents is made to facilitate the extraction of structured data from free-texted inputs, along with the evaluation of the completeness of the data regarding the requirements of the specific country's pharmacovigilance. In contrast to fixed formats, the process dynamically produces minimal follow-up questions to request missing mandatory data.

Supporting documentation like prescription papers or medical receipts can be optionally uploaded to enhance accuracy with automated information extraction like drug name, dosage prescribed etc. When there is a need to clarify medical issues, it is possible to have consent-based follow-up with doctors via secure case identifiers without unsolicited contacts.

All reports, whether full and complete or partial are retained with confidence and completeness scoring to enable audit-ready safety reports. Simultaneously, data is aggregated to enable population-level monitoring to identify spikes and possible safety signals

Figure 1: Problems in Pharmacovigilanc

for medications, symptoms, geography and time.

For the pharmaceutical industry's safety assessment teams, the solution offers a centralized dashboard detailing categorized case summary information, confidence levels, trending, and AI-driven correlation to support and aid effective safety assessment and regulatory decisions. Most notable, however, is that the solution architecture integrates a human-in-the-loop framework to ensure final safety conclusions are within expert, human, control.

On the whole, the proposed solution provides a scalable, user-friendly, and regulatory-compliant methodology for pharmacovigilance follow up modernization through intelligent automation.

WORKFLOW DESCRIPTION

Phase 1: Universal Initiation and Identity Authentication

The reporting journey is designed to be as frictionless as possible to encourage immediate adverse event reporting.

- **Multimodal Entry:** Users initiate reporting via a WhatsApp chatbot using simple text such as "Hi", "", or any greeting.
- **Voice-Assisted Intake:** The chatbot supports voice-to-text input, allowing users to verbally describe their concerns.
- **Dynamic Language Engine:** The system automatically detects the user's language and adapts in real time if the user switches languages during the conversation.
- **Trust Filter:** The system distinguishes between a patient and a healthcare professional (HCP) at the start of the interaction.
- **Verified HCPs:** Healthcare professionals already present in the company registry are immediately trusted and allowed to proceed.
- **New Professional Verification:** New practitioners are required to upload a valid medical license. Verification is completed through a human-in-the-loop process to prevent false clinical claims.
- **Patient Accessibility:** Patient interactions use lay-friendly language and simplified questions to reduce reporting barriers.

Phase 2: Contextual Data Extraction and Localization

Once the reporter identity is confirmed, the system begins automated pre-filling of regulatory requirements.

- **Geo-Compliance:** The reporter's country is identified using the mobile country code.

- **Regulatory Mapping:** The system retrieves the appropriate country-specific pharmacovigilance reporting template from the database.
- **Document Intelligence:** Users may upload prescriptions and medical bills, from which the system extracts drug names, dosages, and clinic details.
- **Minimalist Follow-Up:** Targeted follow-up questions are generated only for information that could not be extracted automatically.

Phase 3: Clinical Triage and Doctor Handover

This phase focuses on evaluating clinical relevance and ensuring medical completeness of the report.

- **Guideline Comparison:** Reported side effects are analyzed against established medicine guidelines.
- **Triage Logic:**
  - *Regular or Known Effects:* The system provides appropriate guidance and reassurance to the patient.
  - *Unusual or New Effects:* The case is treated with high priority and escalated for immediate review.
- **Case ID Handover:** If a report is incomplete, a unique Case ID is generated. The patient may share this ID with their doctor, who can continue the conversation and provide technical medical remarks.

Phase 4: Validation, Analytics, and Global Safety

The final phase transitions from individual case handling to aggregate safety intelligence.

- **Confidence Scoring:** Responses are evaluated for consistency and plausibility, and each case is assigned a confidence score to support prioritization.
- **Case Finalization:** All reports, including incomplete submissions, are retained as structured, audit-ready safety cases with a complete interaction history.
- **Continuous Monitoring:** AI-driven monitoring performs spike detection to identify emerging safety trends across drugs, regions, and time.
- **Safety Dashboard:** Prioritized cases and potential safety signals are presented to pharmacovigilance teams through a centralized dashboard, supported by AI-assisted literature analysis.

OUTCOME

This methodology enables efficient follow-up data collection, improves data completeness and quality, and supports early identification of potential safety risks. By combining regulatory alignment, continuous monitoring, and human-in-the-loop governance, the system ensures transparency, scalability, and compliance across global pharmacovigilance operations.

MODEL CHOICE OVERVIEW

The proposed system follows a task-specific, multi-model design, where different models are used for language understanding, document processing, validation, and signal detection. This modular approach improves robustness, explainability, and regulatory safety compared to a single end-to-end model.

Natural Language Understanding Models

Natural Language Processing (NLP) models are used to process free-text adverse event (AE) reports submitted by patients and healthcare professionals.

Responsibilities:

- Extract drug names, symptoms, severity cues, and timelines
- Identify reporter type (patient vs healthcare professional)
- Normalize medical terminology into structured fields

Training Data:

- Public medical NLP datasets
- Synthetic AE reports generated from pharmacovigilance templates
- De-identified historical safety narratives (where permitted)

Evaluation Metrics:

- Entity extraction accuracy
- Recall on critical safety fields
- Error rate on medical term normalization

Conversational AI & Follow-Up Models

A constrained conversational model is used to drive the chatbot-based follow-up workflow.

Responsibilities:

- Generate minimal, context-aware follow-up questions
- Adapt language and complexity based on user type
- Maintain conversation state across sessions

Key Design Constraint:

- The model operates within predefined templates and rules
- It cannot invent medical advice or change regulatory logic

Evaluation Metrics:

- Follow-up completion rate
- Average number of questions per case
- User drop-off rate

Advanced OCR & Prescription Understanding Models

Specialized OCR and vision-language models are used to process uploaded prescriptions and medical bills.

Responsibilities:

- Read handwritten and printed prescriptions
- Extract medicine name, dosage, frequency, and duration
- Match extracted text against a validated medicine database

Validation Workflow:

- OCR output is cross-checked with a drug reference database
- Low-confidence matches trigger user confirmation

Evaluation Metrics:

- OCR character accuracy rate
- Medicine name matching accuracy
- False extraction rate

Compliance, Scoring & Classification Models

Supervised models assign:

- Completeness score
- Consistency score
- Confidence score

These scores are used for prioritization and routing, not report rejection.

Training Data:

- Simulated incomplete vs complete safety cases
- Rule-generated labels based on regulatory requirements

Evaluation Metrics:

- Precision in identifying incomplete cases
- False escalation rate
- Human reviewer agreement rate

Signal Detection & Population-Level Models

Unsupervised and statistical models monitor aggregated safety data.

Responsibilities:

- Detect frequency spikes
- Identify organ-specific clustering
- Track geographic and temporal deviations

Evaluation Metrics:

- Signal sensitivity
- False-positive alert rate
- Time-to-detection compared to baseline methods

Model Governance & Safety

- All models operate under a human-in-the-loop framework
- Outputs are logged, explainable, and auditable
- No model independently triggers regulatory action

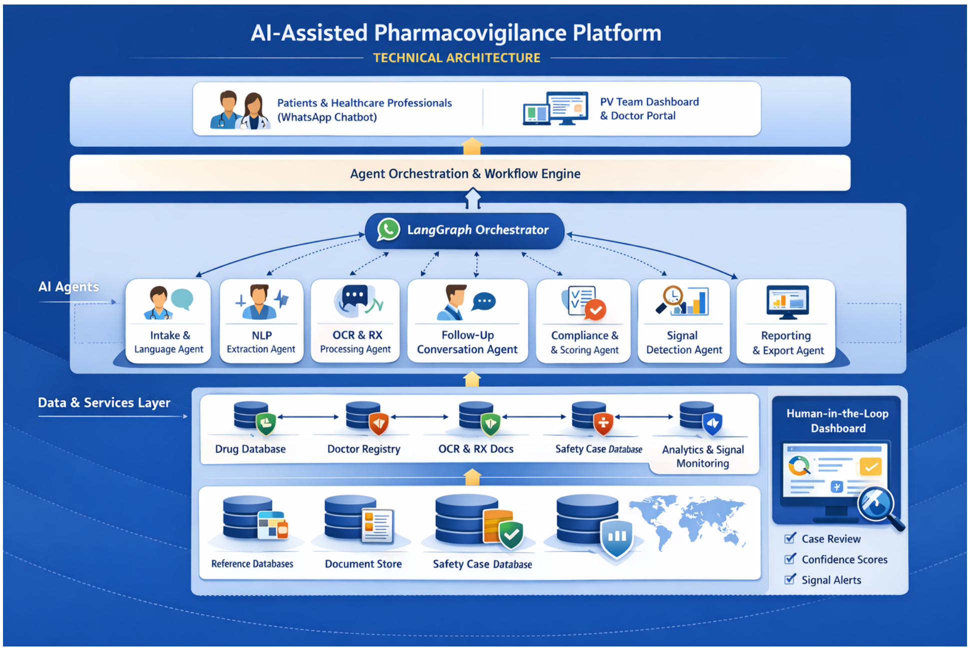

Figure 2: Technical Architecture

TECHNICAL ARCHITECTURE & TECHNOLOGY STACK

System Architecture Overview

The proposed solution is implemented as a modular, event-driven, multi-agent architecture, designed for scalability, auditability, and regulatory safety. The system separates user interaction, AI processing, compliance logic, and analytics into clearly defined layers, enabling flexible deployment and future extensibility.

At its core, the platform uses an agent-orchestrated workflow (LangGraph-based), where each AI agent performs a single, well-scoped task and communicates through structured events. Human oversight is integrated at critical decision points.

High-Level Architecture Layers

6.2.1 Frontend & User Interaction Layer

- WhatsApp Chatbot as the primary user-facing interface for patients and healthcare professionals
- Multilingual conversational flow with consent capture
- Secure web interface for:
  - Doctor follow-up via Case ID
  - Pharmacovigilance team dashboard

Frontend Stack:

- WhatsApp Business API (conversation gateway)
- Web dashboard: React / Next.js
- Authentication: OTP-based + role-based access

6.2.2 Backend & Agent Orchestration Layer

- Central backend service managing workflows, routing, and permissions
- LangGraph-based agent orchestration to control execution order, branching, and retries
- Event-driven communication between agents

Backend Stack:

- Python (FastAPI)
- LangGraph for agent workflows
- Message/event handling via queues (e.g., Kafka / PubSub / SQS)

6.2.3 AI & Agent Layer

Each agent operates independently but within predefined guardrails:

- Language Detection & Intake Agent – identifies language, country, and reporter type
- NLP Understanding Agent – extracts structured AE data from free text
- OCR & Prescription Agent – processes handwritten and printed prescriptions using advanced OCR and vision-language models
- Medicine Matching Agent – validates extracted medicine names against a curated drug database
- Compliance & Completeness Agent – enforces country-specific pharmacovigilance rules
- Follow-Up Strategy Agent – decides if and how follow-ups are required
- Conversational Agent – generates minimal, structured follow-up questions
- Validation & Confidence Agent – checks consistency and assigns confidence scores
- Signal Detection Agent – monitors aggregated data for trends and spikes
- Reporting Agent – formats regulator-ready outputs

AI Stack:

- Transformer-based NLP models
- Vision-language OCR models for handwritten text
- Rule-based engines for compliance
- Statistical & unsupervised models for signal detection

6.2.4 Data Storage & Databases

Safety Case Database:

- Structured adverse event cases
- Version-controlled with full audit trail
- Relational DB (PostgreSQL)

Document Store:

- Secure storage for prescriptions and bills
- Linked to case IDs
- Object storage (S3-compatible)

Reference Databases:

- Drug & medicine database (validated names, formulations)
- Company-maintained healthcare professional registry (where applicable)
- Country-specific pharmacovigilance templates

Analytics Store:

- Aggregated, anonymized data for signal detection
- Columnar or time-series database

6.2.5 Human-in-the-Loop Dashboard

A secure dashboard for pharmacovigilance teams enables:

- Case review and prioritization
- Confidence and completeness visualization
- Safety signal alerts
- Manual overrides and escalation
- Literature review and supporting evidence analysis

6.2.6 Security, Privacy & Compliance

- Explicit consent capture at intake
- Data encryption at rest and in transit
- Role-based access control
- No unsolicited doctor outreach
- Full audit logs for regulatory review
- Separation of conversational interface and data storage

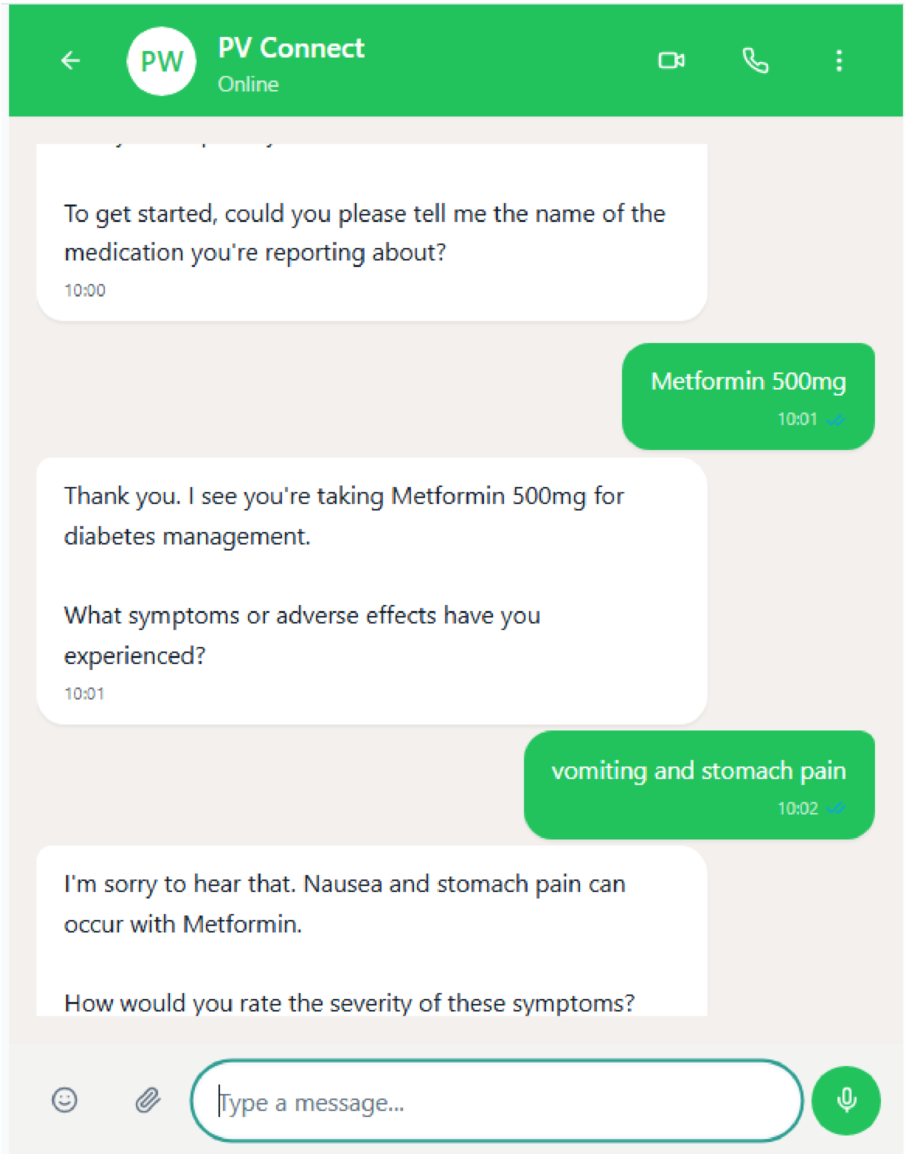

Figure 3: CHATBOT INTERFACE

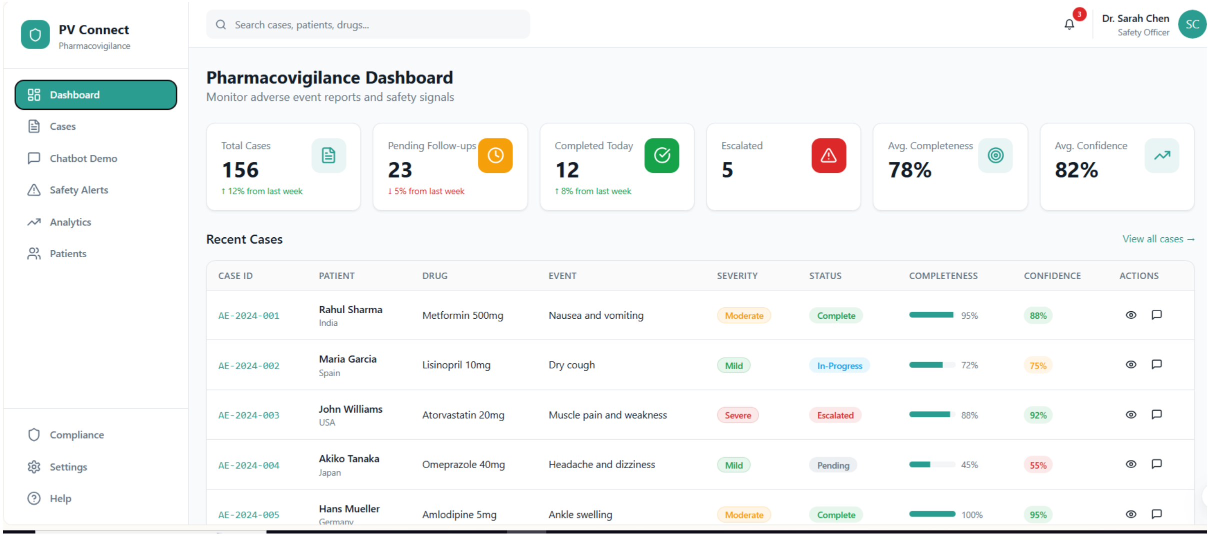

Figure 4: Dashboard Idea

CHALLENGES AND NEXT STEPS

Although the proposed solution is designed to align closely with real-world pharmacovigilance workflows, certain challenges are expected during implementation. Patients and healthcare professionals may sometimes provide incomplete information or disengage during follow-up due to time constraints or situational pressures. Differences in regulatory requirements across countries also introduce complexity in maintaining consistent compliance logic at scale. In addition, handwritten prescriptions can be difficult to interpret accurately despite advanced OCR models, and safety signal detection may occasionally identify trends influenced by external factors such as seasonal illnesses or increased public awareness rather than true drug-related risks. Managing the workload of pharmacovigilance teams during periods of increased reporting is another practical consideration.

Moving forward, the next steps include refining and validating models using de-identified real-world data, continuously updating regulatory rule sets, and improving human-in-the-loop workflows through smarter prioritization. Future work will focus on enhancing analytics and visualizations and conducting pilot deployments with pharmacovigilance teams to gather feedback and progressively improve system usability, accuracy, and integration into existing safety operations.